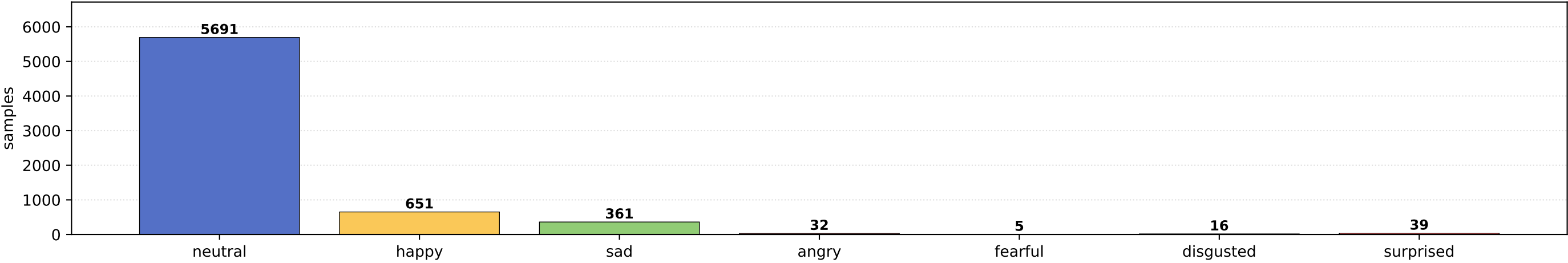


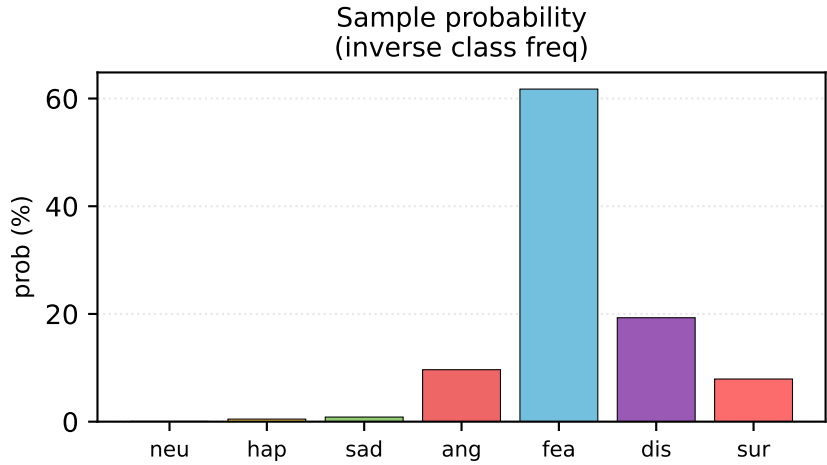
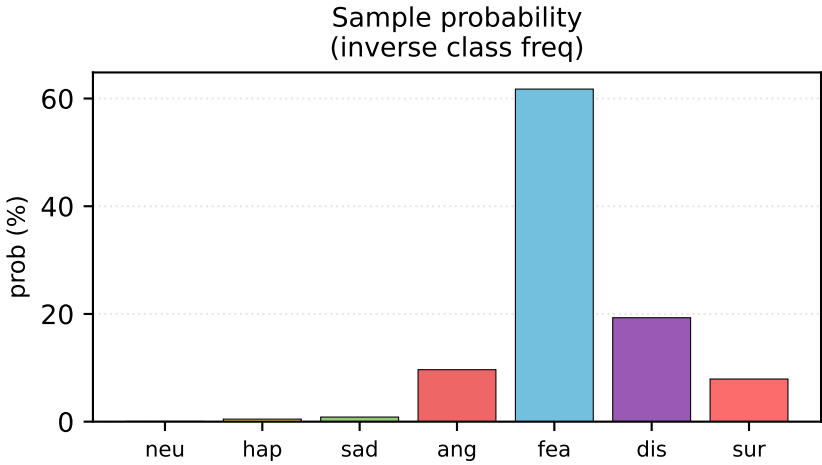
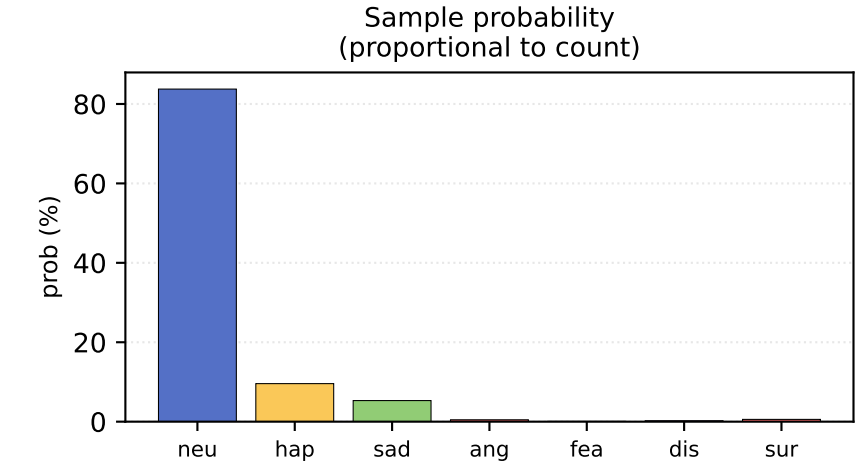
Sampler Strategy — B1 / B2 / B3 (3 imbalance scenarios)
Class distribution Primer — max:min ratio = 1138:1 (severe imbalance)



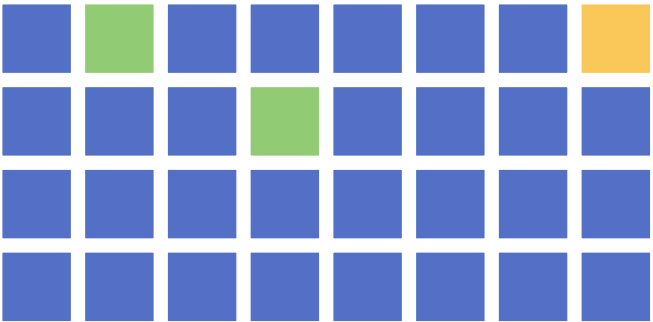
B1
Shuffle uniform sampler
+ no augmentation

B2
WeightedRandomSampler
(prob $\propto$ 1/class_count)
+ no augmentation

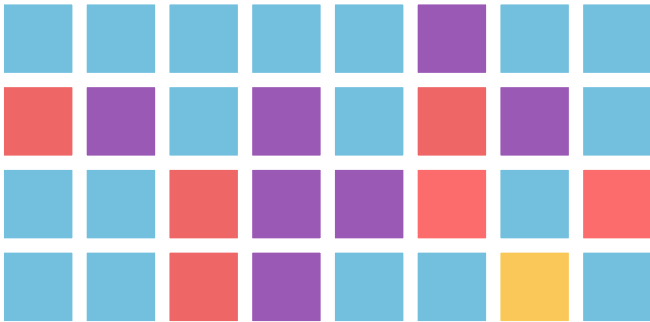
B3
WeightedRandomSampler
+ synced per-batch aug
(hflip + rotate + photo)



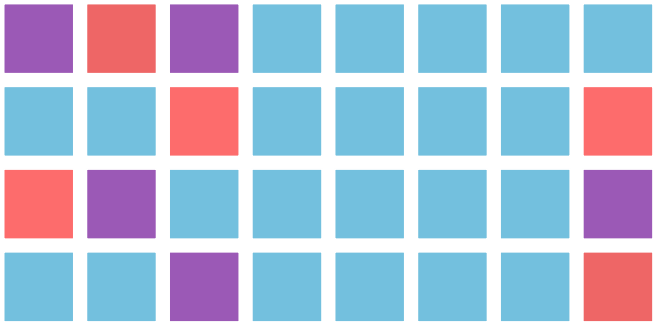
Simulated batch (32 samples)
neu=29 hap=1 sad=2



Simulated batch (32 samples)
hap=1 ang=4 fea=18 dis=7 sur=2



Simulated batch (32 samples)
ang=2 fea=22 dis=5 sur=3



✗ No augmentation
(raw sample direct)

✗ No augmentation
(raw sample direct)

✓ Synced per-batch aug:
• hflip + landmark swap + heatmap flip
• rotate $\pm 10^\circ$
• brightness/contrast $\pm 10\%$

neutral happy sad angry fearful disgusted surprised